

SFWRENG 4HC3/6HC3
Human-Computer Interfaces
Department of Computing and Software
Fall 2024

Course Information

Lecture: TuFr 2:30PM – 3:20PM (PGCLL M21)
Th 5:30PM – 6:20PM (CNH 104)

Tutorials: Th 12:30PM – 1:20PM (T01, PGCLL M21)
Tu 1:30PM – 2:20PM (T02, PGCLL M21)
We 2:30PM – 3:20PM (T03, KTH 104)
Tu 4:30PM – 5:20PM (T04, IWC 223)
Mo 10:30AM – 11:20AM (T05, BSB 104)

Final Exam: Final exam will be scheduled during the final examination week
(Friday, December 6 to Thursday, December 19)

Instruction Team Information

Instructor: **Irene Ye Yuan**, Ph.D., irene.yuan@mcmaster.ca
Office Hours: TuTh 9:30AM – 11:00AM
ABB C529/over Zoom by request
Additional times available by appointment

TA Information: **Natasha Atokolo**, atokolon@mcmaster.ca
Habib Ghaffari-Hadigheh, ghaffh1@mcmaster.ca
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Note: Make sure you include the course title and number in the email subject line [SFWRENG 4HC3/6HC3]. If you have general question about the course content, make sure you use the appropriate Teams channels to post your questions. It may take **up to**

one day for us to reply to emails and posts during weekdays, and **up to two days** during the weekend.

Course Description

Design of user interfaces. Principles of good interface design. Task-oriented design. User experience design. Inclusive design. Communicating with graphics. Modes and mode awareness problem. Human cognition (memory, perception, motor systems). Help systems. Interface design tools. Three lectures, one tutorial (one hour). One term.

Prerequisite(s): Registration in Level III or above of any Software Engineering program, or registration in an Integrated Biomedical Engineering and Health Sciences (IBEHS) program and either COMPENG 2SI4 or permission of the department.

Antirequisite(s): COMPSCI 4HC3, SFWRENG 4D03

Cross-list(s): COMPSCI 4HC3, SFWRENG 4HC3, SFWRENG 6HC3

Course Objectives

This course introduces you to human-centered design. You will develop a vocabulary for evaluating interface designs from the perspective of their intended users. You will apply this knowledge to interaction designs for your choice of user goals and/or tasks based on data gathered from your intended users. You will evaluate at least one of your designs with user evaluation techniques.

By the end of this course students will be able to:

1. Analyze interaction and interface design solutions from a critical perspective including fundamental design principles, human needs, and capabilities.
2. Apply design research methods and data analysis skills to inform and guide interaction and interface design processes.
3. Develop user-focused requirements, task analysis, scenarios, and storyboards to visualize and plan user interactions with the interface.
4. Produce prototypes with appropriate fidelities and iterate on design solutions based on user feedback and testing

Books, Software, and Supplies Required for the Course

- **Textbook:** Don Norman. 2013. *The Design of Everyday Things*. Basic Books; Revised edition, 2013 ([available as an eBook from the McMaster library](#)).
 - Additional readings from will be provided and posted to Avenue.
- **Online Software:** This course does not teach how to use specific software packages, and there is no requirement about the use of specific software. This is up to students to decide. In addition to packages and tools students are already familiar with, this course will also introduce students to a variety of design and prototyping software (e.g., Figma, Marvel, Axure).

- **Other Supplies:** Please **always** bring a few sheets of **paper** and **pencils/pens** (or something for digital drawing) to the class since most classes will involve some kind of sketching. You are welcome to use online tools and/or tablets for these in-class exercises as well. Please also make sure you have a smartphone with a camera you can use to take photos of what you've sketched, upload, and share if you use paper (please try to keep the phone away from you otherwise during the class).

Graded Components

Individual Assignments (25%)

There will be two individual assignments during this course. The assignment details will be posted on Avenue, and the assignments will be submitted through Avenue. Students enrolled in 6HC3 will have additional components required for the second assignment.

- **Assignment 1 (10%):** Interaction Critical Evaluations
- **Assignment 2 (15%):** Interface Design and Prototyping Exercise

Group Project and Team Evaluations (45%)

The course places a large emphasis on group-based work based on a multi-phase design project. Students are required to form a project team about 5 members. Please note that all group members must be registered in the same lab section. The group project is cumulative, consisting of multiple project deliverables. It is essential that each deliverable is submitted on time. A list of specific due dates and deliverables will be posted to Avenue. Here is a tentative list of the four project milestones:

- **Milestone 1 (4%):** Proposal & Team Contract
- **Milestone 2 (16%):** User Requirements Elicitation & Definition
- **Milestone 3 (17%):** Design Sketches & Low-Fidelity Prototype
- **Milestone 4 (8%):** High-Fidelity Prototype & Project Report

Group Project Team Evaluations (5%)

At Milestone 2 and Milestone 4, you will be asked to provide feedback and evaluation for your team members. Specific instruction and rubric will be provided for completing evaluation and providing feedback. Here is a list of breakdowns for these two project milestone team evaluations:

- **Milestone 2 (3%):** Mid-Project Team Evaluation
- **Milestone 4 (2%):** Final-Project Team Evaluation

In-Class Activities (5%)

5 in-class activities will be administered during the lecture time (2% x 5 in total). These activities help your practice skills and content from lecture and prepare you better for your assignments and design project. In-class activities are expected to be completed during the class time but may need extra time to wrap up after the class (often due at midnight on the same day of the class). Each activity will be graded on **an effort-based system** (2 points in total): 1 point for completing the activity, and 1 point for demonstrating good effort in the submitted activity.

Final Exam (20%)

There will be one final exam for this course during the scheduled final exam period. The exam will be written and closed book exams (no electronic aids), but you may bring five double-sided cheat sheets (8.5x11) to the exams.

Component Weights

Component	Weight
2 Individual Assignments	25%
4 Group Project Milestones	45%
2 Group Evaluations	5%
5 In-class Activities	5%
1 Final Exam	20%

Grading Distribution

The following breakdowns will be used when assigning final grades. Any adjustment will be made to ensure grades are assigned fairly.

- 90%-100%: A+
- 85%-89%: A
- 80%-84%: A-
- 77%-79%: B+
- 73%-76%: B
- 70%-72%: B-
- 67%-69%: C+
- 63%-66%: C
- 60%-62%: C-
- 57%-59%: D+
- 53%-56%: D
- 50%-52%: D-
- 0%-49%: F

Tentative Class Schedule

Week	Dates	Topic	Tutorials	Due
1	9/3-9/6	Introduction to HCI	(No tutorial)	
2	9/10-9/13	Design Principles	Design Project M1 work time	
3	9/17-9/20	User Requirements Gathering	Design Project M1 check-in (project pitch)	Design Project M1
4	9/24-9/27	User Requirements Gathering	Design Project M2 work time	
5	10/1-10/4	Human Abilities	M2 In-Tutorial Pilot	Assignment #1 check-in
6	10/8-10/11	Human Abilities	Design Project M2 work time	Assignment #1
7	10/15-10/18	Mid-term Recess	(No tutorial)	
8	10/22-10/25	Inclusive Design	Design Project M2 work time	Design Project M2
9	10/29-11/1	Prototyping	Design Project M3 work time	
10	11/5-11/8	Prototyping	Design Project M3 check-in	Assignment #2 check-in
11	11/12-11/15	Evaluation	M3 In-Tutorial Evaluation	Assignment #2
12	11/19-11/22	Evaluation	Design Project M3 check-in	Design Project M3
13	11/26-11/29	Evaluation	Design Project M4 work time	
14	12/3-12/1	Course wrap-up	(No tutorial)	Design Project M4
	12/6-12/19			

Class Policies

Late Submission Policy

There is a late-days system for this course as things might happen from time to time. You will have a total of **6 late days** for individual assignments and a total of **15 late days** (per group) for project milestone deliverables. We will use Avenue to track your remaining late days. Some assignments may have limits on how many late days you can use. Be mindful about how many late days you want to use for your assignments, as there might not be enough time for us to provide feedback for the following assignments. After the late days, there will be a **10% penalty per day** (up to a max of 50%). Submissions that receive **more than 50% penalty** will be marked as 0.

Course Etiquette

Lectures are **automatically scheduled for recording** and recordings **will be posted on Avenue after the class** (it may take several hours to process the video). However, the recordings are not guaranteed (as technology may fail sometimes). **Do attend lectures and participate in the in-class activities** and use lecture recordings as **your last resort** when you have emergent situations or conflicts.

Students are expected to treat their fellow students in the class, the instructor, and the teaching assistants in **a respectful and professional manner**. **No personal audio or video recording** of lectures or presentations is allowed in any format otherwise, openly or surreptitiously, unless requested by additional accommodation. Course materials (both paper and digital) are for the students' **private study only**. Posting, sharing, or selling course material (either online or in-person) is a violation of copyright law and academic integrity.

MSAF Policy

Any exam, assignments, or project submissions missed due to MSAF **will be deferred to a later date**. Failure to arrange and complete the deferred exam or submission will result in a mark of 0.

Using Generative AI in Course Assessments

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to...submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also state in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative

AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity. ¹

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¹ <https://mi.mcmaster.ca/generative-artificial-intelligence-in-teaching-and-learning/#tab-content-provisional-guidelines>

COURSE OUTLINE – APPROVED ADVISORY STATEMENTS

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/), located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other

students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).²

² Updated to the *Policy on Requests for Relief for Missed Academic Term Work* effective May 1, 2021

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email

**This syllabus is adapted from versions created by previous SFWRENG/COMPSCI 4HC3/6HC3 instructors*